

Telegram Keyword Search

Multi-User Bot & Desktop Application

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Title:	Telegram Bot Developer Python Automation & Multi-User Systems
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Project Type: Telegram Bot & Desktop Application - Multi-User Chat Search System

Lines of Code	Platforms	Commands	Status
1,200+	2 (Bot + Desktop)	8 Commands	Production Ready

■ Technology Stack

Category	Technologies
Bot Framework	Python 3.11+, Telethon (MTPROTO API), python-telegram-bot
Desktop Application	Tkinter (Native GUI), PyInstaller (Executable)
Authentication & Security	Per-user sessions, 2FA support, Encrypted storage, .env config
Backend & Processing	Async/await (asyncio), Date filtering, Regex matching, Multi-chat aggregation
Development & Deployment	Multi-process architecture, Modular structure, Git version control

■ Project Overview

A dual-interface Telegram chat search system that empowers users to find specific messages across their Telegram history using keywords and date filters. Built with Telethon for direct Telegram API access and python-telegram-bot for bot interface, supporting both a Telegram bot and a native Windows desktop application. Features per-user authentication where each user logs in with their own Telegram account (not the bot owner's), ensuring privacy and security. Implements advanced search capabilities including multi-keyword queries, date range filtering, and smart chat selection. The system uses secure session management with persistent storage, allowing users to maintain login state between sessions. Desktop app built with Tkinter provides a native GUI experience with clickable message links, while the Telegram bot offers in-chat convenience. Designed for professionals managing large communication histories, researchers analyzing conversation patterns, or anyone needing fast retrieval of specific information from thousands of messages. Supports MTPROTO proxy for restricted regions and includes comprehensive error handling with user-friendly feedback.

■ Key Features

- 1. Dual Interface Design** - Available as both Telegram bot (use from mobile/desktop Telegram) and standalone Windows desktop app with Tkinter GUI, offering flexibility for different user preferences.
- 2. Per-User Authentication System** - Each user logs in with their own Telegram account credentials (phone + verification code + optional 2FA), ensuring complete privacy where users search their own chats.
- 3. Multi-Keyword Search** - Search for multiple keywords simultaneously (comma-separated), finding messages containing any or all specified terms across selected chats.
- 4. Date Range Filtering** - Specify start and end dates (YYYY-MM-DD format) to narrow searches to specific time periods, perfect for retrieving conversations from specific projects or timeframes.
- 5. Flexible Chat Selection** - Choose to search specific chats/groups by name (comma-separated) or search all chats at once, with automatic exclusion of 'Saved Messages' and bot chats for cleaner results.
- 6. Secure Session Management** - User sessions stored locally in encrypted files, persisting login state between bot restarts while allowing instant logout and session deletion via /logout command.

7. **Desktop Executable (.exe)** - Includes PyInstaller configuration to build standalone Windows executable, requiring no Python installation for end users, making it easy to distribute.
8. **Simultaneous Dual Operation** - Unique multi-process architecture allows running both Telegram bot and desktop GUI simultaneously, serving users on both platforms concurrently.
9. **Direct Message Links** - Search results include clickable Telegram message links (tg://openmessage?) that instantly navigate to the exact message in the chat, saving time scrolling through histories.
10. **MTPROTO Proxy Support** - Built-in support for MTPROTO proxies to bypass Telegram restrictions in countries with limited access, ensuring global usability with configurable proxy settings.

■ Application Screenshots

Telegram Keyword Search

Not logged in

Login Search Results

Step 1: Enter Phone Number

Phone (international format):

+

Send Code

Step 2: Enter Verification Code

Code from Telegram:

Verify Code

Step 3: Two-Factor Authentication (if enabled)

2FA Password:

Verify 2FA

Logout

Please log in to search your chats

Desktop GUI Login Screen - Tkinter interface showing phone number input, verification code field, and 2FA password field with clean modern design.

Telegram Keyword Search

Telegram Keyword Search

Logged in as: A m i r

Login

Search

Results

Chats & Folders to Search

Enter 'all', chat names, or folders with <folder_name> syntax:
Examples: all | <Work>, <Friends> | chat1, chat2, <MyFolder>

<Graphic>

Show My Folders

Exclude (Optional)

Exclude chats/folders using the same syntax (<folder_name>, chat):
Examples: <Muted>, Ads, <SpamFolder> | leave empty for none

پروژه های پارسکدرز

Keywords

Keywords (comma-separated):

python

Date Range

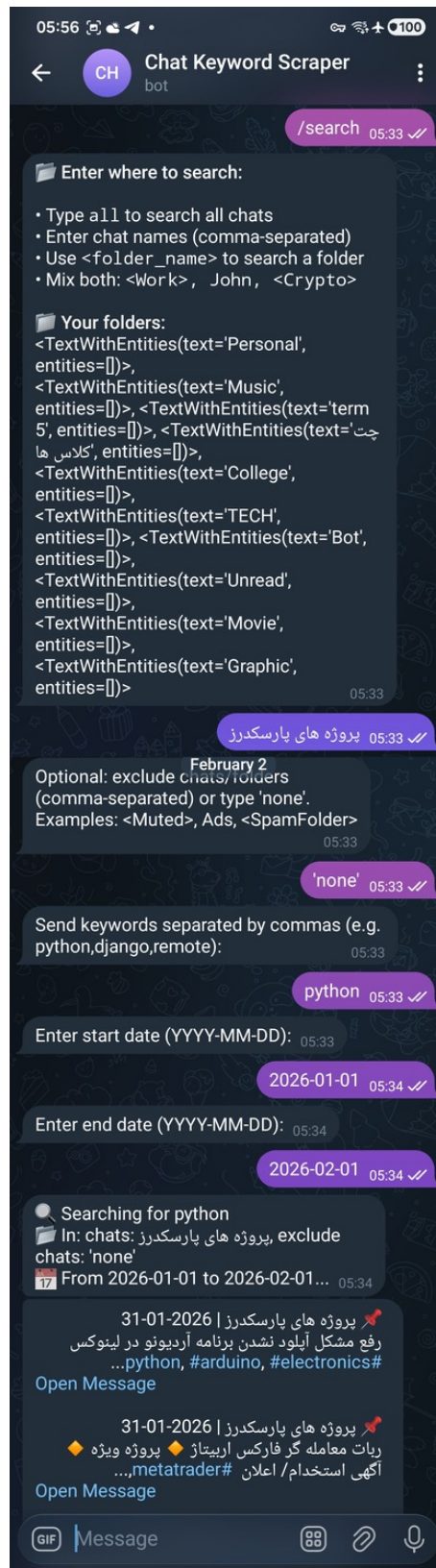
Start Date (YYYY-MM-DD):
2026-02-01

End Date (YYYY-MM-DD):
2026-02-02

Search Messages

Found 12 messages

Desktop Search Interface - Main app window with chat selection, keyword input fields (comma-separated), date range pickers, and search results area.



Telegram Bot Conversation - Mobile screenshot showing bot interaction: /start welcome message with inline keyboard and /login flow with phone entry.

Telegram Keyword Search

Logged in as: AAmiirA

Login Search Results

Found 12 messages (showing up to 50)

- 20:38 01-02-2026 | پروژه‌های کارلنسر**
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 ...تب صفحه مشاهده برای @ ۱۴۰۴ بهمن ۱۲ یکشنبه :انتشار تاریخ 📅 روز ۱۰
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Search Results Display (GUI) - Desktop app results showing formatted message excerpts with timestamps, chat names, and clickable message links.



Search Results Display (Bot) - Telegram bot results with formatted message excerpts, timestamps, and clickable tg:// message links.

■ Skills Demonstrated

Category	Skills
Telegram API Development	Telethon MTProto API integration • Multi-user session management • 2FA handling • Bot command handlers with conversation states • Inline keyboards and reply markup design
Desktop Application	Tkinter GUI design and layout management • Cross-thread communication (asyncio + Tkinter) • Native Windows application building • PyInstaller executable packaging • Session persistence
Python Backend	Async/await with asyncio • Multi-process application design • Modular package structure • Shared utility modules • Environment variable configuration • Exception handling and logging
Authentication & Security	Per-user authentication (phone + verification + 2FA) • Secure session token storage • Session lifecycle management • Privacy-first architecture • Credentials management with .env
Search & Data Processing	Keyword matching across multiple chats • Date range filtering with timezone awareness • Chat history pagination • Regex pattern matching • Result aggregation • Performance optimization

■ Similar Projects I Can Build

- **Telegram Automation Bots:** Multi-user reminder/scheduling bots • File management bots (search, organize, backup) • Chat statistics analyzers (message counts, activity patterns) • Auto-reply bots with customizable rules • Group management bots (permissions, moderation)
- **Desktop Applications with Bot Integration:** Telegram backup tools with GUI (export chats, media) • Cross-platform messaging dashboards (aggregate multiple accounts) • Contact synchronization tools • Media downloader apps with progress tracking • Chat migration tools between platforms
- **Search & Retrieval Systems:** Email search tools with keyword/date/sender filtering • Document search engines for local file systems • Social media content aggregators with search capabilities • Log file analyzers with pattern matching • Database query interfaces with natural language processing
- **Authentication & Multi-User Systems:** OAuth integration tools for third-party API access • Session management systems for web applications • User permission management with role-based access control • API gateway with per-user rate limiting • Secure credential storage systems with encryption
- **Dual-Interface Applications:** CLI + GUI combinations for developer tools • Web interface + desktop app for data management • Mobile-responsive web apps with Telegram bot fallback • API + GUI admin panels for backend systems • Voice interface + visual dashboard combinations

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Telegram Bot Developer | Python Automation & Multi-User Systems

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